Kamiak Cluster Architecture

How the machines behind `ssh kamiak` fit together, and what that means for your jobs

146	7,136	20	1 PB	100 Gb/s
compute nodes	CPU cores	GPUs on 6 nodes	all-flash WekaIO storage	InfiniBand HDR100

1 The whole system

Kamiak is a *condominium* cluster. Faculty investors and colleges buy compute nodes; WSU provides the head nodes, the network, the storage and the staff. Every machine plugs into one InfiniBand fabric, so a file written anywhere appears everywhere, and a job can spread across nodes. Figure 1 shows each of the 146 compute nodes as one slab, grouped by the partition that owns it.

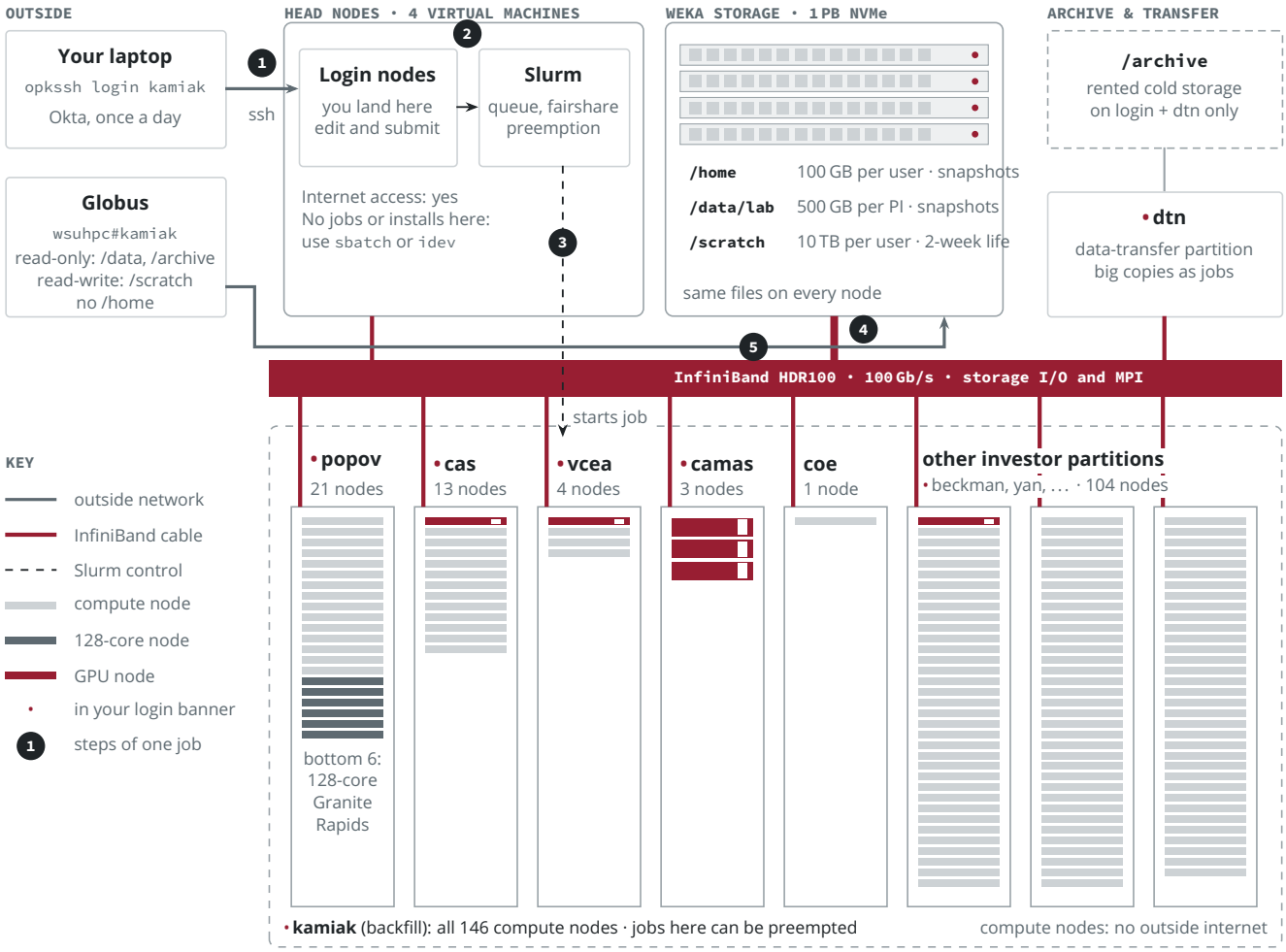


Figure 1. Kamiak's hardware and the path of one job (**1**–**5**, detailed in Figure 6). Racks are grouped by partition, not by physical position. The number of Weka storage servers is illustrative; CIRC does not publish it.

2 Compute nodes

The nodes span ten years of Intel Xeon generations. Most *nodes* are old, but most *cores* are new: the 50 Sapphire Rapids and Granite Rapids nodes, the generations `popov` is built from, hold 57% of the cluster's cores.

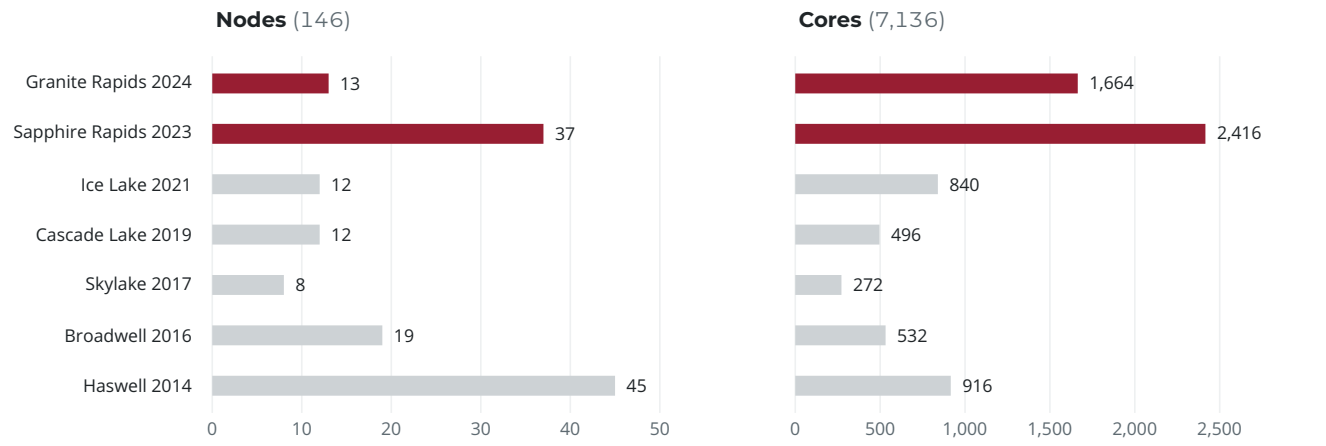


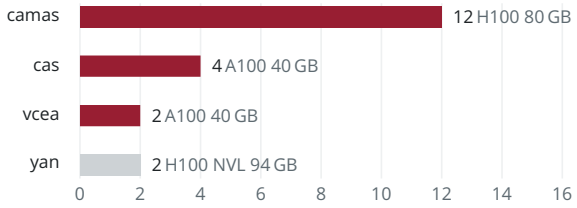
Figure 2. Kamiak by CPU generation (year of the Xeon generation's launch). Crimson marks the two newest generations: 34% of nodes, 57% of cores. Haswell is the reverse: 31% of nodes, 13% of cores. Source: hpc.wsu.edu partition list.

Inside `popov`

Nodes	Cores	RAM*	CPU
cn154–168 (15)	64	100 GB	Sapphire Rapids
cn177–182 (6)	128	151 GB	Granite Rapids

*Memory Slurm lets a job request, from `sinfo`, June 2026. A 64-core, ≤100 GB request can start on all 21 nodes; a 128-core request only on 6.

GPUs



GPUs per partition. Crimson: partitions in your banner. Only `camas` is open to everyone, and GPU jobs there are never preempted.

3 Partitions are labels on nodes

A partition is not a separate machine; it is a list of nodes. The same `popov` node is also a `kamiak` node, because the backfill partition covers every node on the cluster (Figure 3). That overlap is what makes `kamiak` free to use, and it is also why jobs there can be killed.

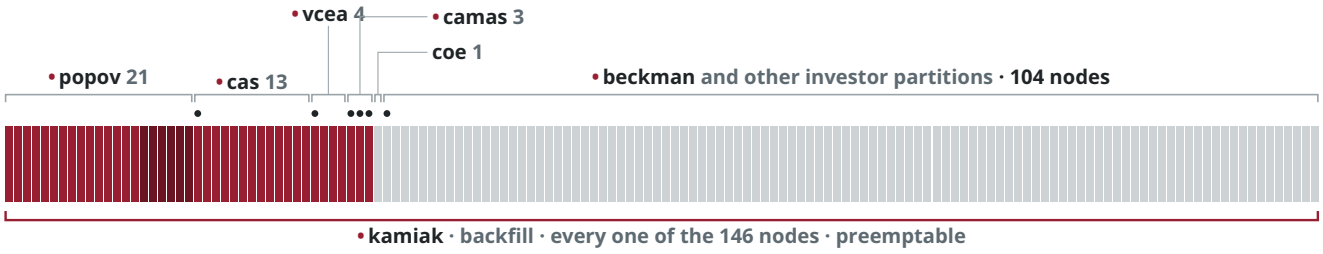


Figure 3. All 146 compute nodes in a row. Crimson: partitions in your banner (darker: `popov`'s 128-core nodes). Dots: GPU nodes. Every node sits in its owner's partition *and* in `kamiak`. `dtn` (data transfer) is not drawn; its nodes are not published.

What preemption does to one node

When an owner needs cores that a backfill job holds, Slurm cancels the backfill job and puts it back in the `kamiak` queue. The work done since the last restart file is lost.

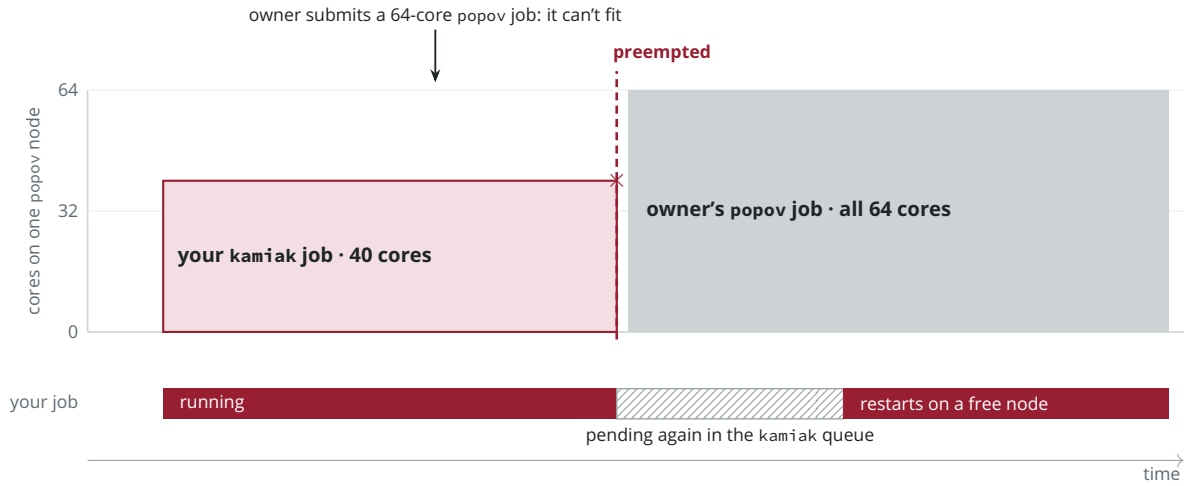


Figure 4. Preemption on one 64-core popov node. The node does not change; only who holds its cores. A job that writes restart files resumes where it left off; one that doesn't starts over.

4 Storage: one filesystem, many servers

`/home`, `/data/lab` and `/scratch` are not on the login nodes or the compute nodes. They live on WekaFS, a parallel filesystem: each file is cut into chunks spread across many flash servers, with extra protection chunks so a failed drive or server loses nothing. Every node mounts it over InfiniBand, so it behaves like one local disk that every node shares.

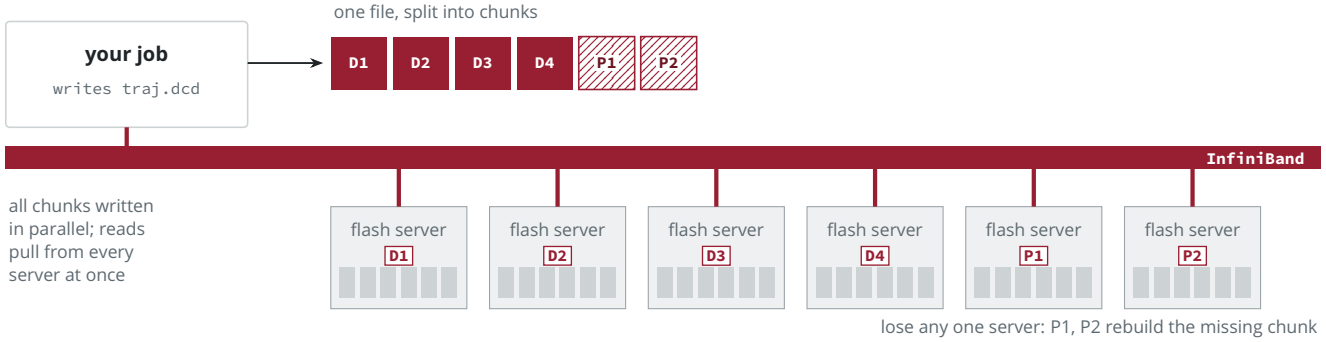


Figure 5. How WekaFS stores one file (illustrative: Kamiak's chunk and server counts are not published). Large reads and writes use every server at once; millions of tiny files are slower, because each one costs a metadata lookup.

Path	Size	Kept	On compute nodes
`/home`	100 GB per user	daily snapshots, kept 3 days	yes
`/data/lab`	500 GB per PI, more for rent	daily snapshots, kept 3 days	yes
`/scratch`	10 TB per user	workspaces deleted after 2 weeks (`mkworkspace`)	yes
`/archive`	rented	cold data	**no**: login and `dtn` only

Table 1. Storage tiers. All share the same flash and the same speed; they differ in size and lifetime. Nothing is backed up off the cluster.

5 One job, start to finish

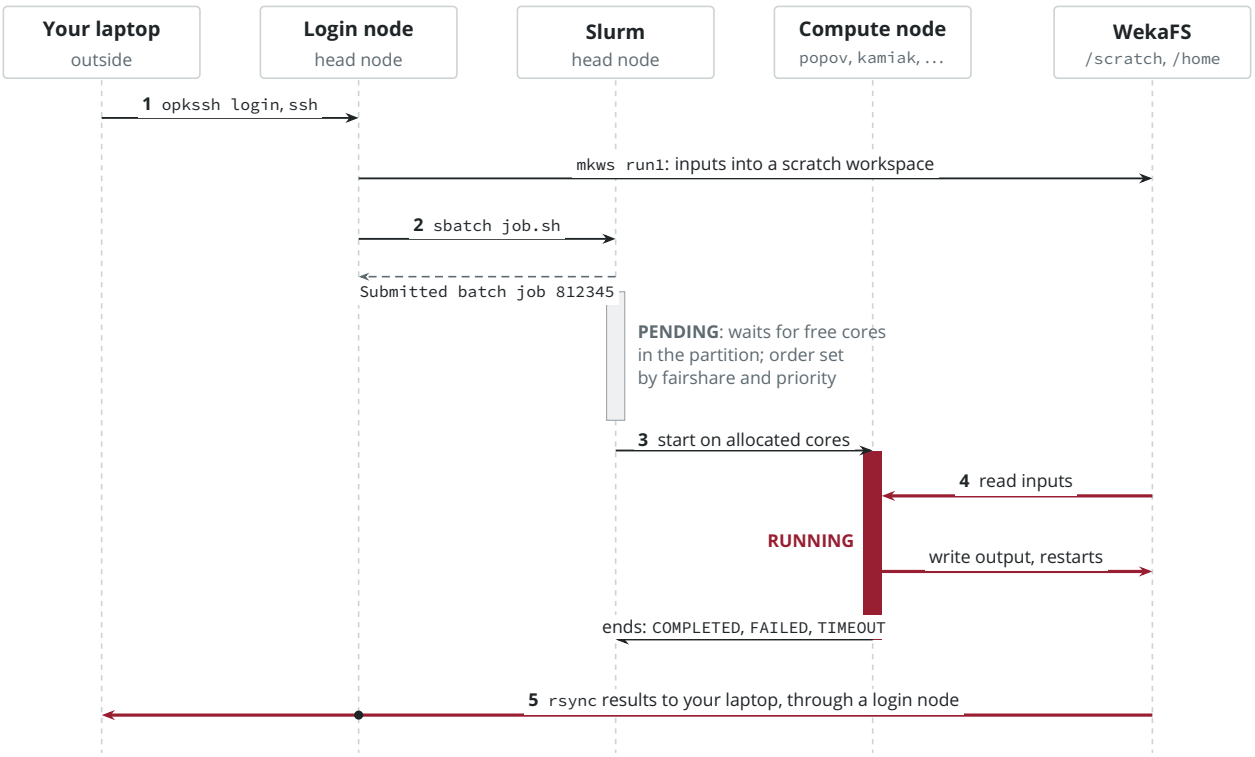


Figure 6. A batch job over time, top to bottom. Crimson arrows carry your data; black arrows are commands and scheduler events. Steps **1**–**5** match Figure 1.

6 Your partitions

Partition	Kind	Nodes	Preempted?	Notes
`kamiak`	backfill, open to all	146	**yes**	every node; use for short or checkpointed runs
`popov`	investor	21	no	no `--exclusive`; ask for cores and `--mem` instead
`camas`	NSF GPU, open to all	3 (12 H100)	no, for GPU jobs	non-GPU jobs are moved to `kamiak`; cite NSF award CNS-2216108
`cas`	college	13 (4 A100)	no	College of Arts and Sciences
`vcea`	college	4 (2 A100)	no	Voiland College of Engineering and Architecture
`beckman`	investor	not published	no	size not published; `scontrol show partition` lists it
`dtn`	data transfer	not published		large copies, and the only compute access to `/archive`

Table 2. The partitions in your login banner. You can list several in one job, e.g. `#SBATCH -p popov,kamiak`; Slurm starts it on whichever frees up first.

7 What this means for your research

- **Run long jobs where they can't be killed.** Multi-day CP2K AIMD and umbrella-sampling windows belong on `popov`, not `kamiak`. Anything on `kamiak` should write restart files often (Figure 4).
- **Size jobs to the hardware.** 64 cores and $\leq$100 GB fits all 21 `popov` nodes and starts soonest. For 64–100 atom cells, one node is the sweet spot; spreading across nodes adds InfiniBand traffic that rarely pays at that size.
- **Use arrays for many similar runs.** Dozens of PMF windows or LAMMPS replicas go in one job array, as the login

banner asks.

- **Train ML potentials on `camas`.** Twelve H100s, open to all and not preempted for GPU jobs. Acknowledge NSF CNS-2216108 in papers that use them.
- **Download before you submit.** Compute nodes have no internet: fetch pseudopotentials, basis sets, model weights and packages on a login node first.
- **Keep files big and few.** WekaFS streams large trajectories fast; millions of small files are slow. Tar outputs, and keep conda environments out of `/scratch`.
- **Move results out of `/scratch` within two weeks**, to `/data/lab`, and keep a copy off the cluster.
- **Methods sections.** Report the CPU generation (Sapphire or Granite Rapids for `popov`), the HDR100 interconnect, and the CIRC acknowledgement: *"This research used resources of the Center for Institutional Research Computing at Washington State University."*

Sources (read 12 October 2026): hpc.wsu.edu pages *Cluster Resources* (node, core, GPU, head-node, network and storage figures), *Storage*, *File Transfers*, *Quick Start Guide*, *Camas GPU Cluster* and *Acceptable Use Policy*; `sinfo -p popov -N -l` (June 2026). Not published, and therefore not drawn: the management Ethernet network, the InfiniBand switch layout, physical rack positions, Weka server count, and node counts for investor partitions other than `popov`. Colours: WSU crimson #981E32 and gray #5E6A71.